Evaluating Resource Adequacy Using Hourly 40 Weather-Year Data Under High VRE Penetration for the Luzon, Visayas, and Mindanao Grids in Alignment with the Power Development Plan(2023-2050)

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The Problem

- **PDP targets:** 35% renewable by 2030, 50% by 2040
- Current planning does not account for **multi-year extreme weather patterns**
- Wind and solar can drop significantly for **extended periods** during challenging years
- **Risk:** Systematic underestimation of capacity needed for supply adequacy
- International studies show single weather-year analysis leads to **misrepresentation of investment needs**

Luzon Power Demand and Supply Outlook

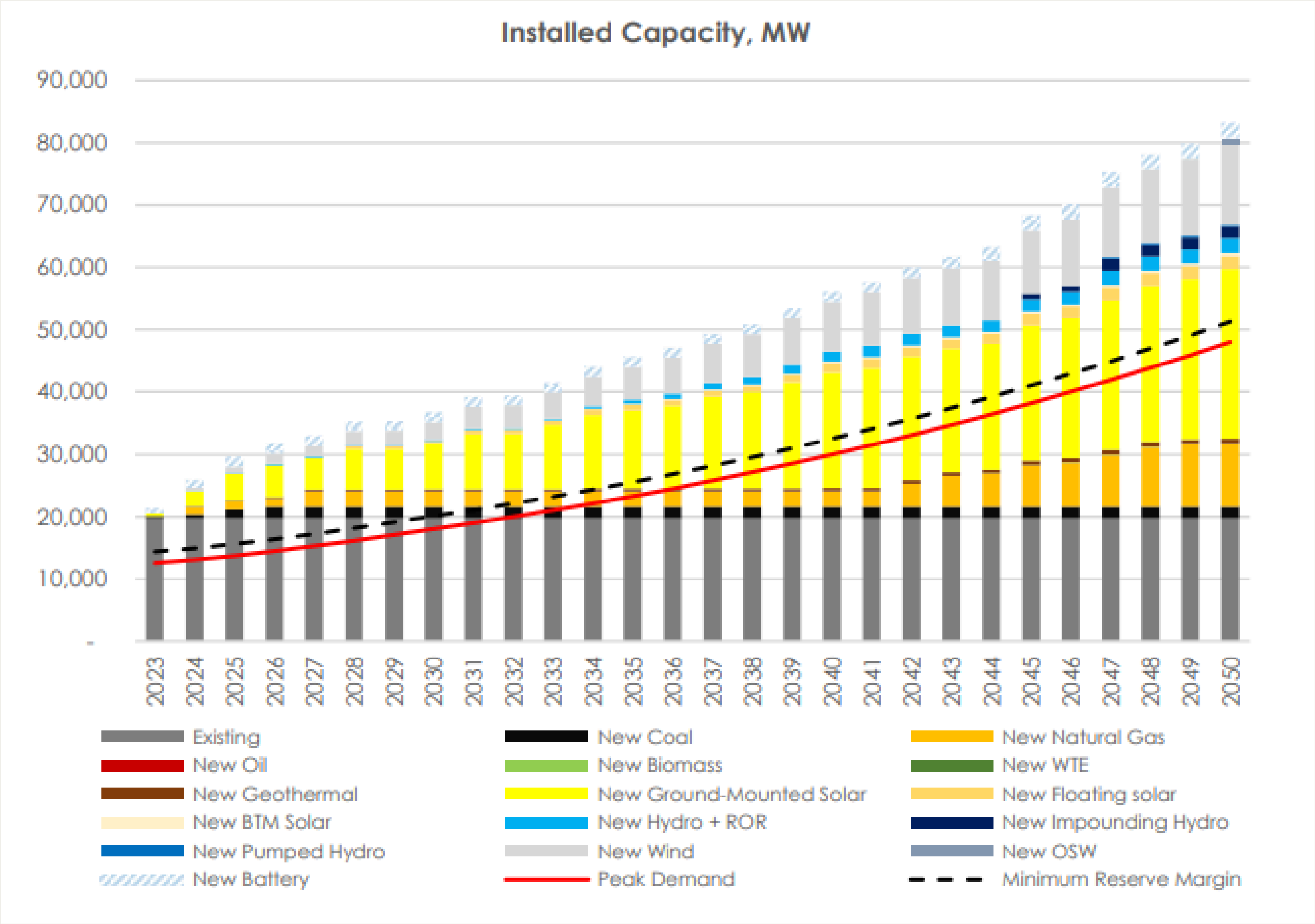


Figure 1.1: Reference Scenario

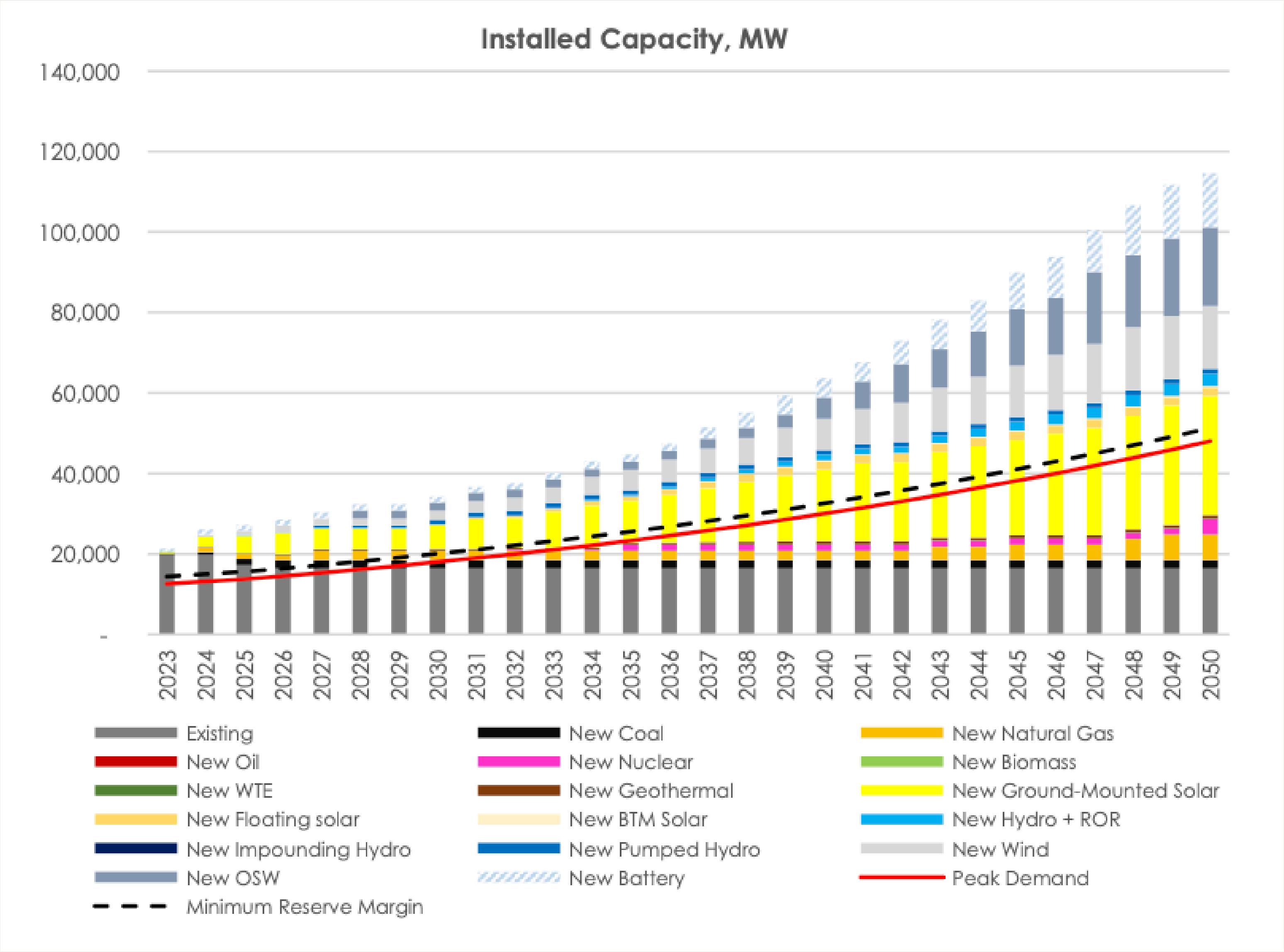
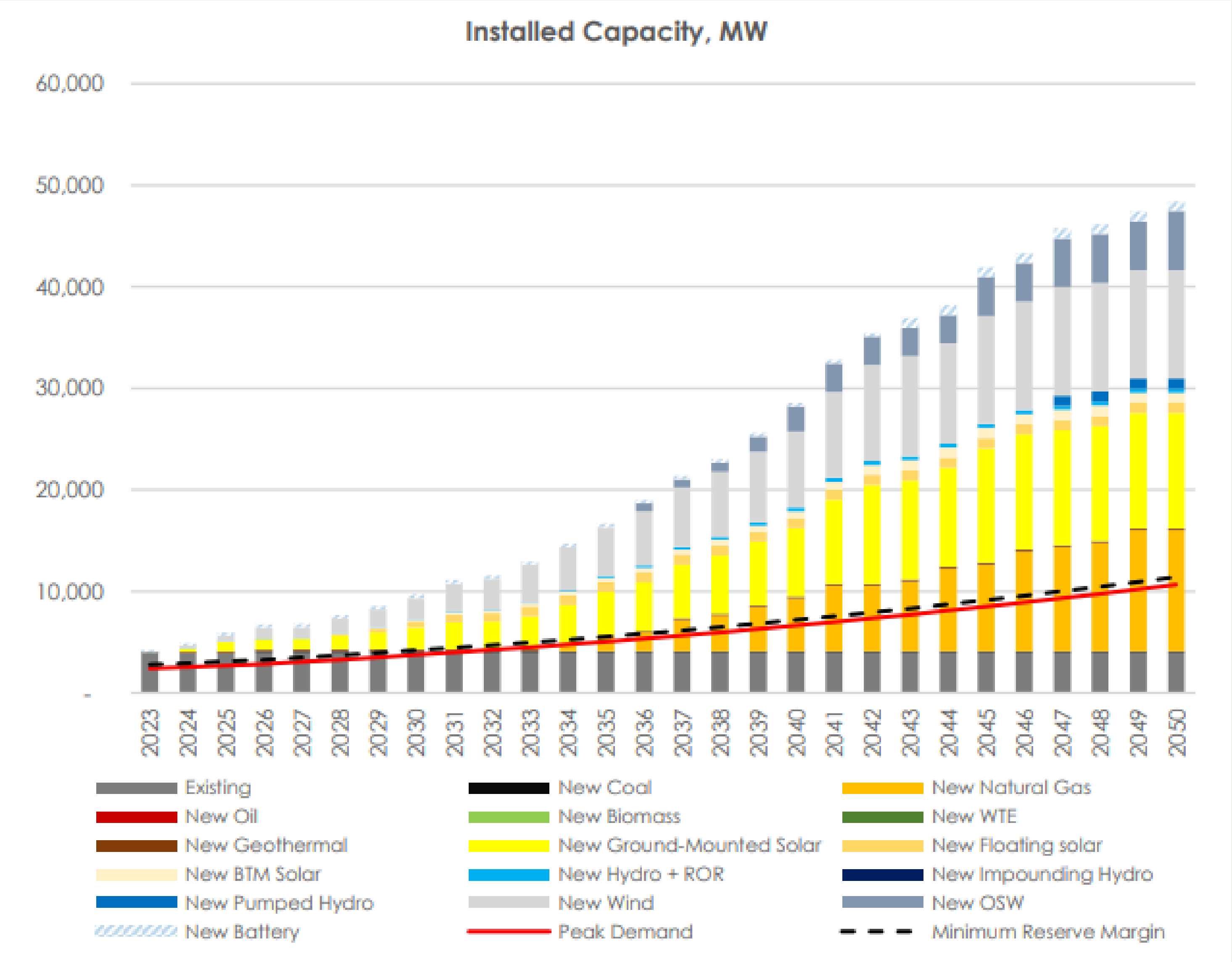


Figure 1.2: CES 1

Visayas Power Demand and Supply Outlook



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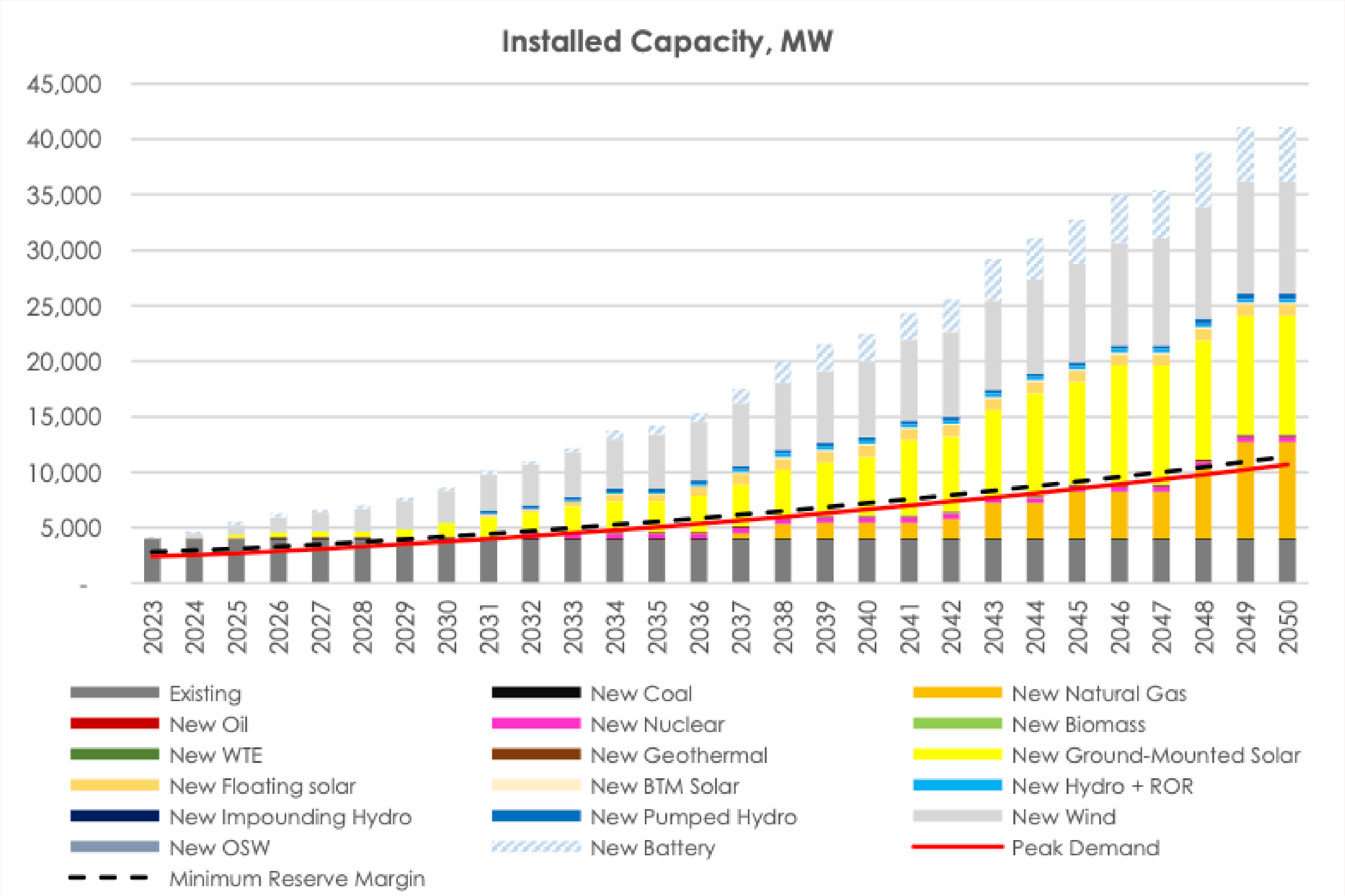

Figure 1.2: CES 1

Mindanao Power Demand and Supply Outlook

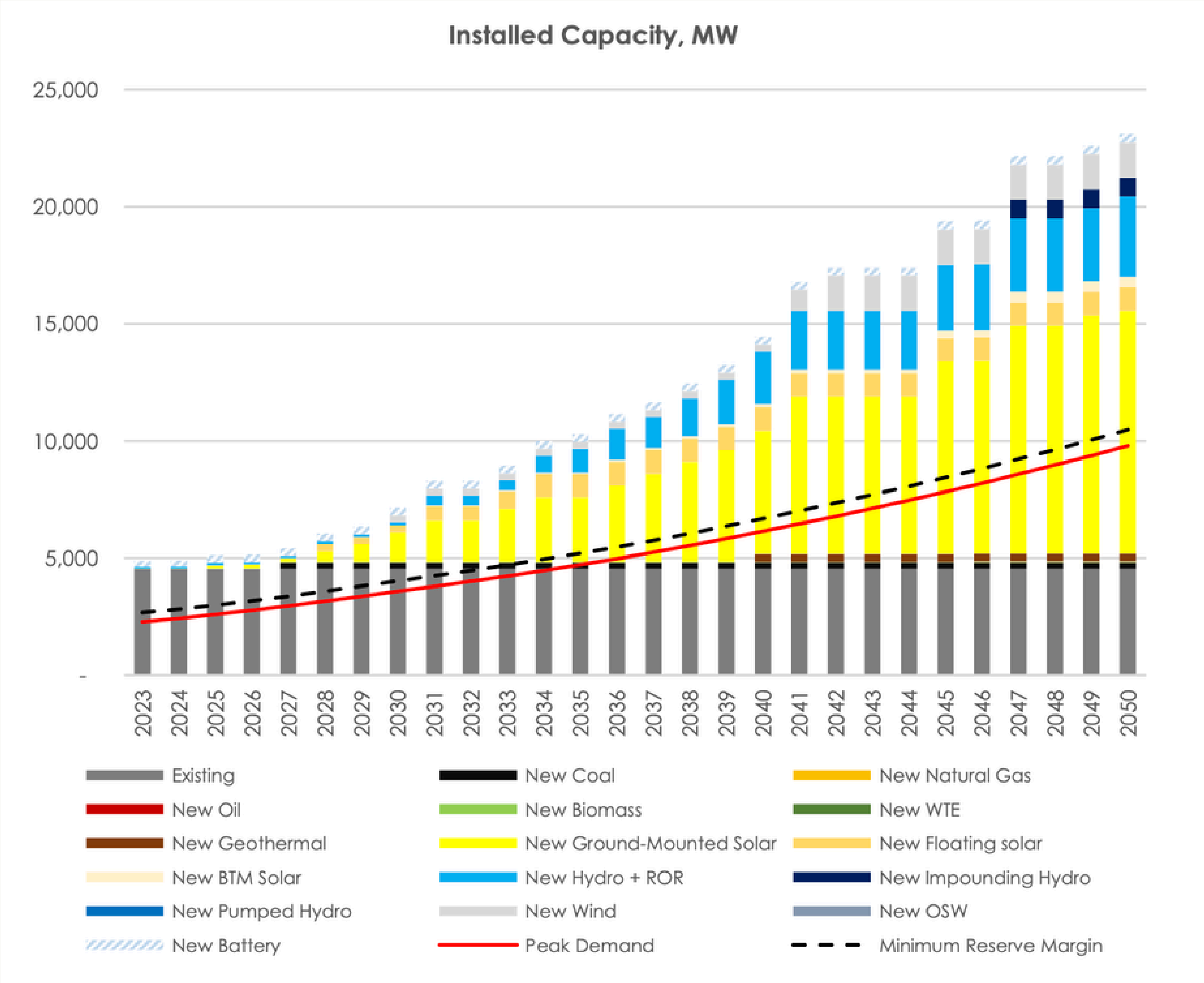

Figure 1.1: Reference Scenario

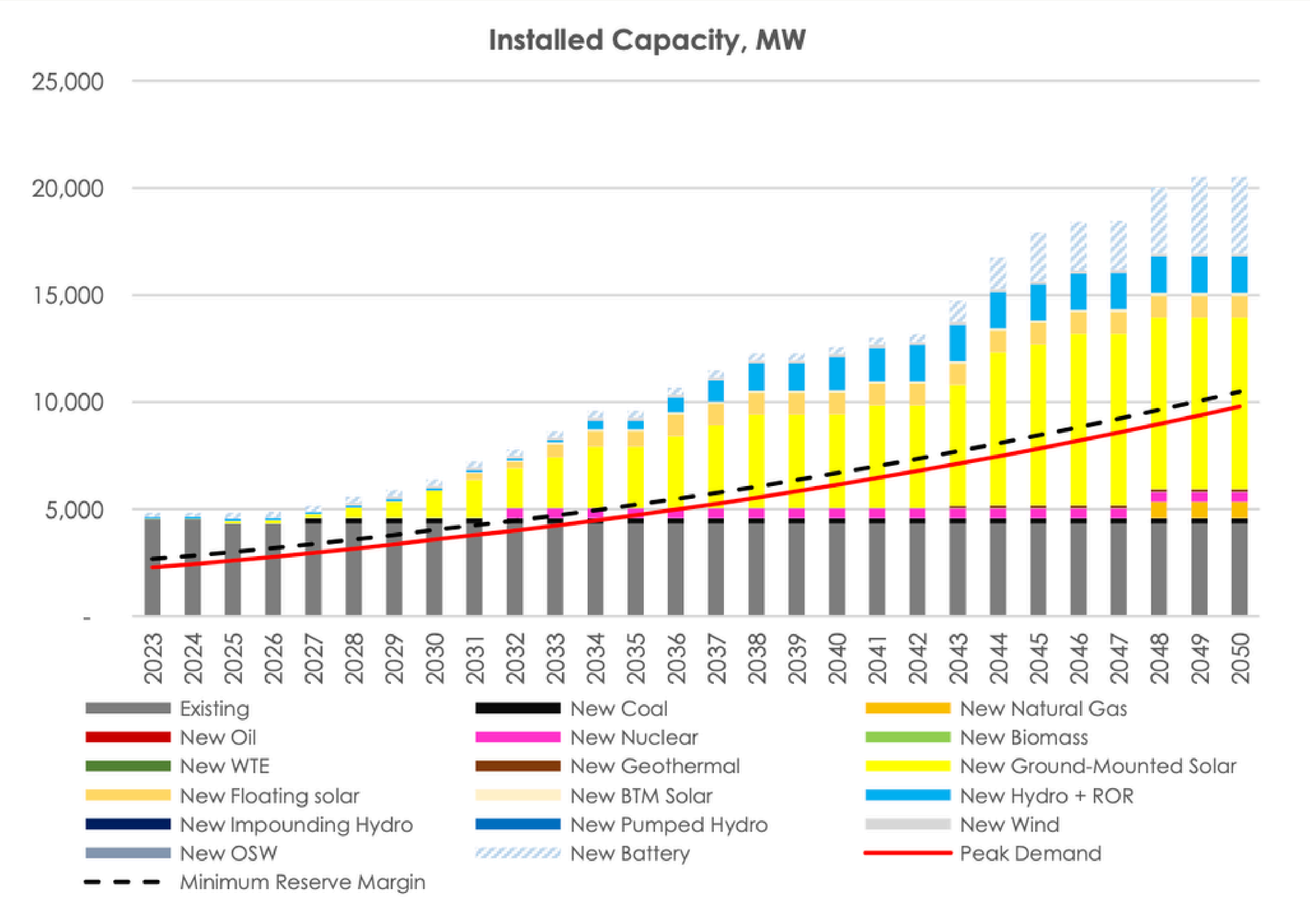

Figure 1.2: CES 1

Resource Adequacy

- **Definition:** Ability of supply-side and demand-side resources to meet aggregate electrical demand at all times
- **Metrics:** Loss of Load Expectation (LOLE), Expected Unserved Energy (EUE), etc.
- **Critical for:** Ensuring reliable service while balancing economic capacity investments

Resource Adequacy

Table 1: Existing Studies on Resource Adequacy

Study	Focus	Tools	Method	Results
A Probabilistic Method to Assess Power Supply Adequacy for the Pacific Northwest [18]	Pacific Northwest adequacy standard	Genesys	Probabilistic LOLP with Monte Carlo; critical water conditions	5% LOLP threshold established
New Resource Adequacy Criteria for the Energy Transition: Modernizing Reliability Requirements [17]	Modernizing adequacy criteria for energy transition	Conceptual framework	Multi-metric (LOLE, EUE); stress-testing extreme scenarios; cost reliability trade-offs	LOLE insufficient alone; EUE preferred for energy-constrained systems; tail risks require deterministic testing.
Representing net load variability in electricity system capacity expansion models accounting for challenging weather-years [7]	Weather-year selection for capacity expansion	ECE model	Net-load recurrence matrix fingerprinting; iterative optimization; handpicked extremes + optimized typical years	4–6 representative years capture 39-year net-load variability; extreme events drive thermal and storage investment.

Visualization

Capacity Risk Zone

- The right of the vertical axis (red) will be called Capacity Risk Zone where the PDP lacks in dependable capacities.

Energy Risk Zone

- The values above the curve (orange) is the Energy Risk Zone where the batteries cannot handle such long-term events.

Safe Zone

- The area between the curve and the vertical axis (green) are the events that the dependable capacity targets can handle easily.

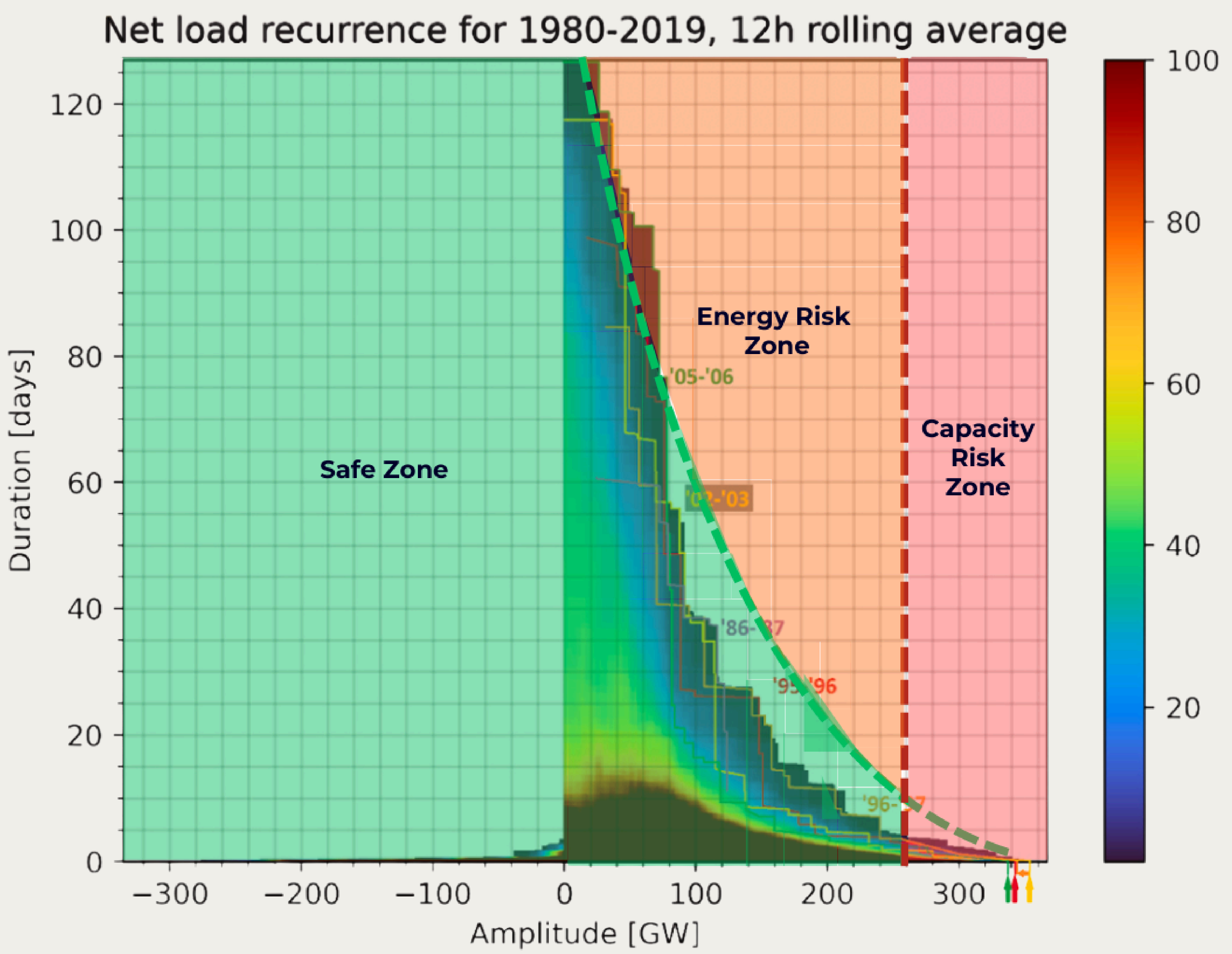

Figure 7: Figure 3 from Ullmark et al [7] with "Capacity Risk Zone", "Energy Risk Zone", and "Safe Zone" Determined

Project Objectives

The general objective of this study is to assess the sufficiency of the generation capacities of the Philippine power system under the capacity targets of the PDP 2023-2050 by characterizing the interannual variability of renewable energy resources. Specifically, this study aims to:

a. Generate multi-decadal solar/wind profiles (40 years: 1985-2024)
b. Characterize net load variability across all CREZ zones
c. Evaluate adequacy of PDP projected capacities
d. Visualize reliability risk zones on recurrence heatmaps

Scope and Limitations

Scope

- Luzon, Visayas, and Mindanao grids
- 2025-2050 (5-year intervals)
- 2 PDP scenarios (Reference Scenario and CES 1)

Limitations

- Single-node assumption (no transmission losses/congestion)
- MERRA-2 historical data only (no future climate change modeling)
- Spatial aggregation at CREZ centroids
- Battery storage: 4-hour fixed duration assumption

Methodology

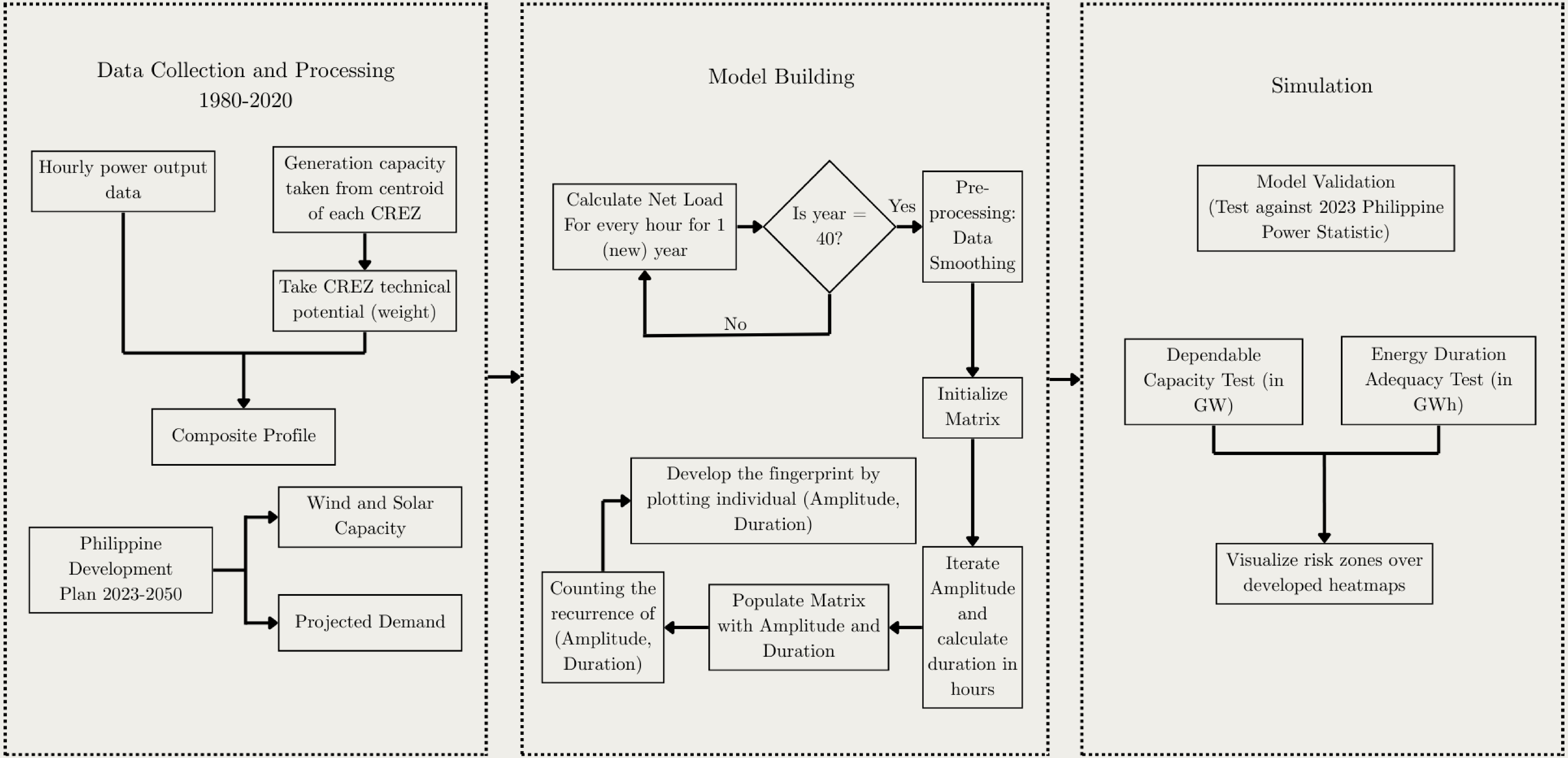

Figure 3: Methodology Overview Flowchart

Methodology: Data Collection

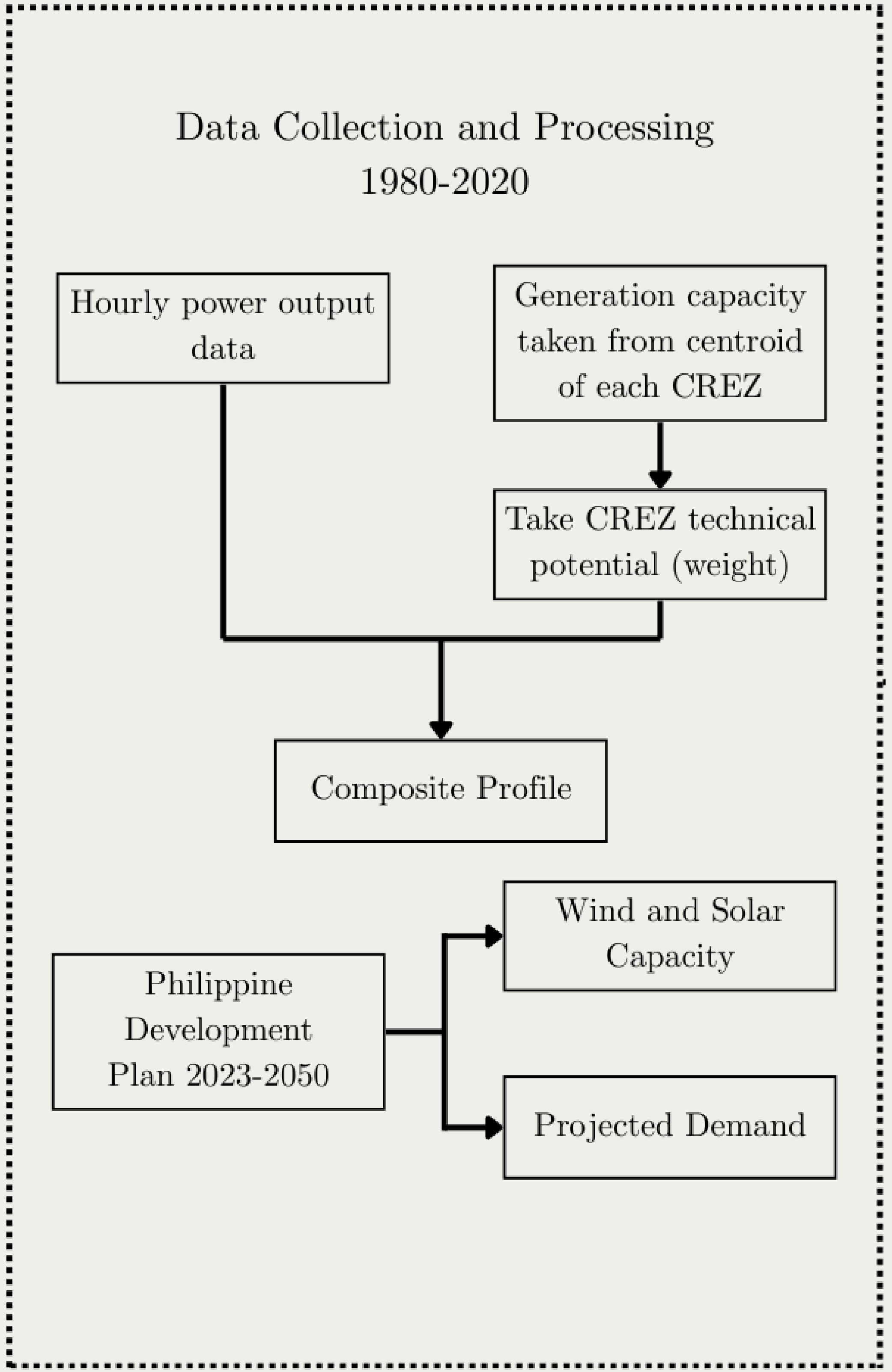

Figure 4: Data collection and Processing

Power Development Plan (2023-2050)

- **Scenarios:** Reference Scenario and Clean Energy Scenario 1 (CES 1)
- **Metrics:** Generation Capacity per technology and Peak Demand Forecast

Renewable Energy Data from Renewables.ninja

- **Historical Data:** Hourly power output from 1985-2024
- Uses MERRA-2 dataset

Competitive Renewable Energy Zones

- Uses centroid of CREZ as inputs

Data Collection: Luzon PDP Scenarios

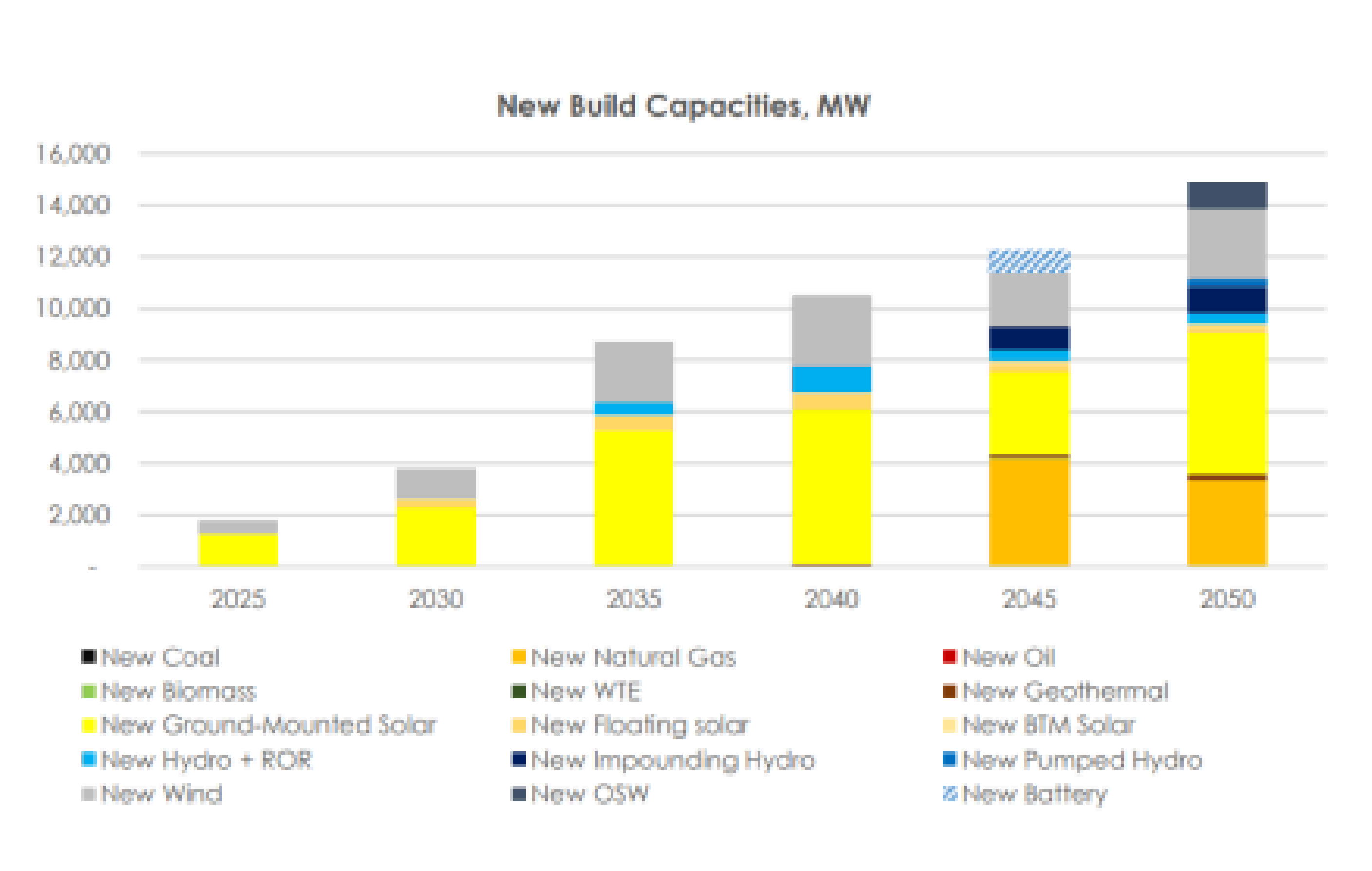

Figure 5.1: Reference - Luzon

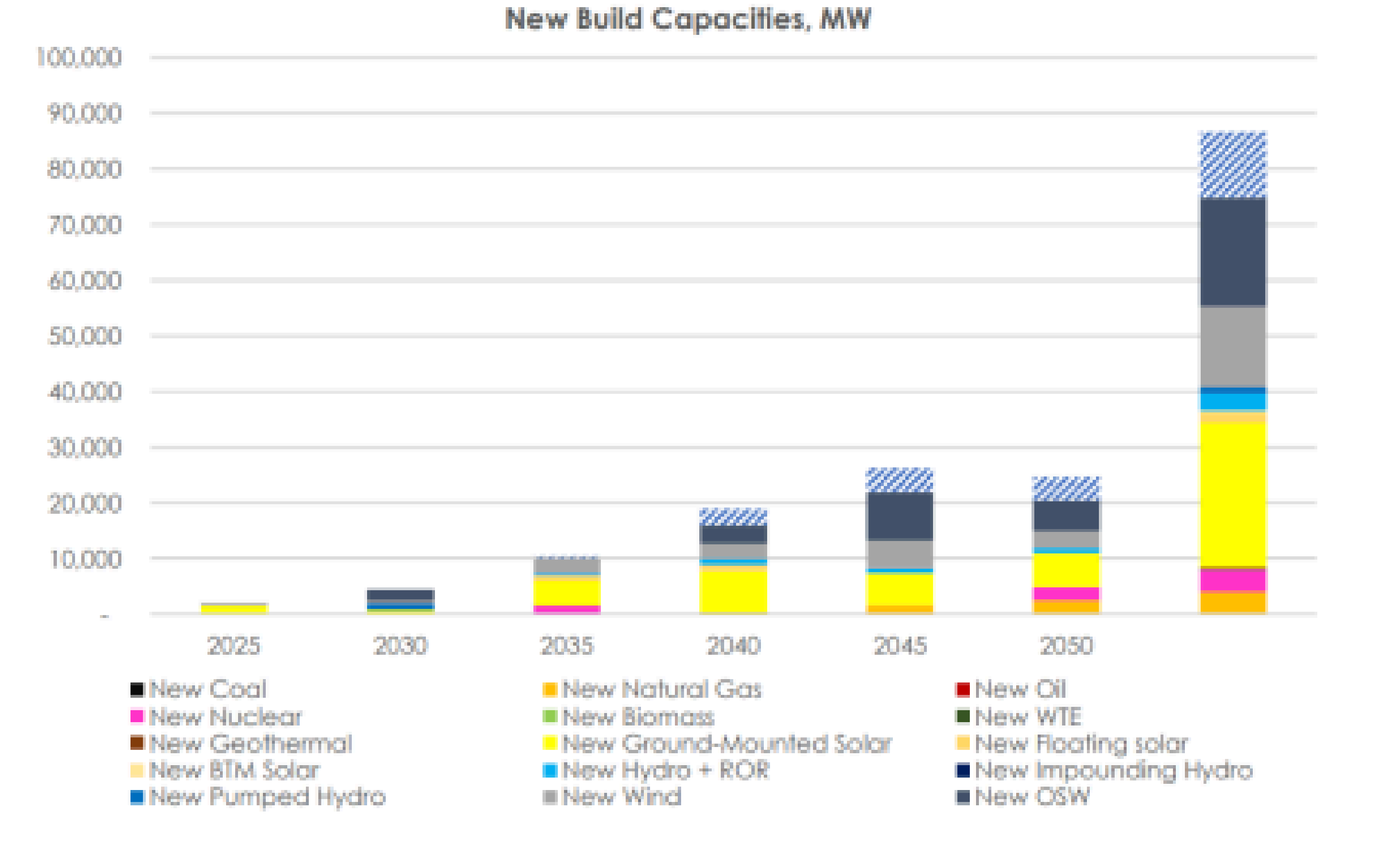

Figure 5.2: CES 1 - Luzon

Data Collection: Visayas PDP Scenarios

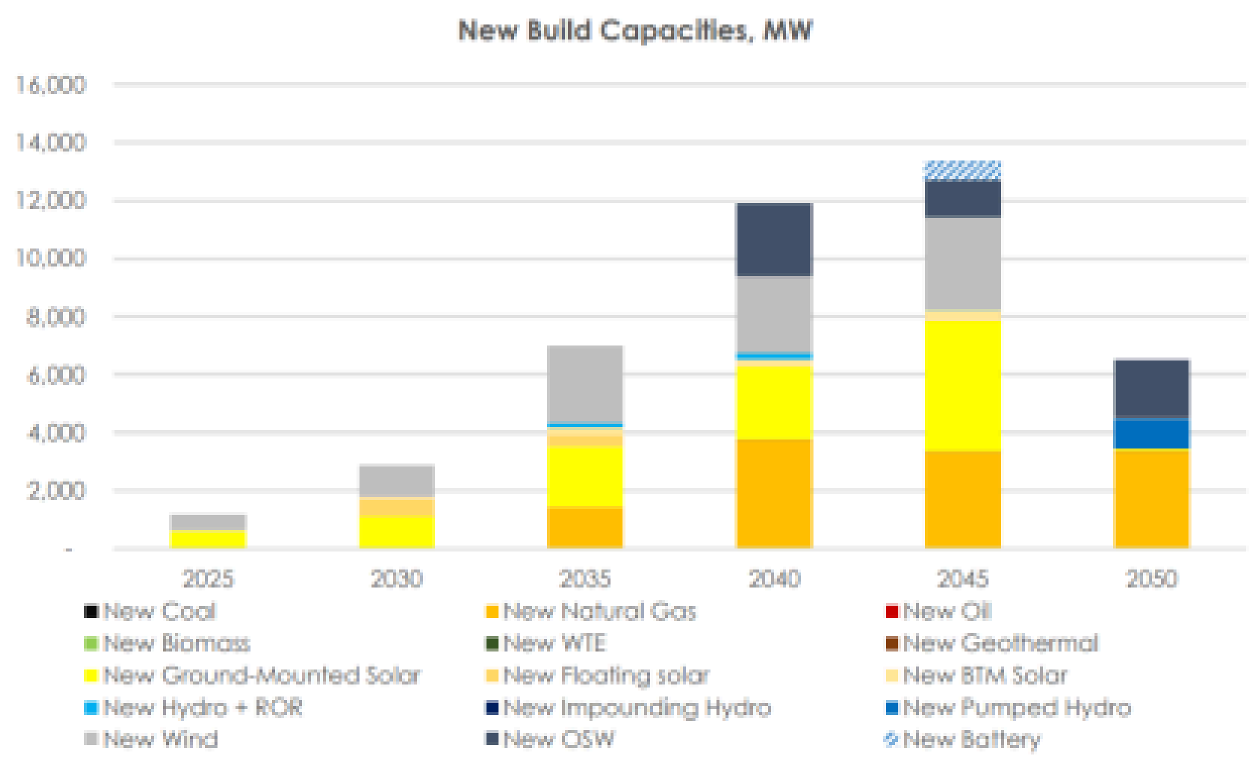

Figure 6.1: Reference - Visayas

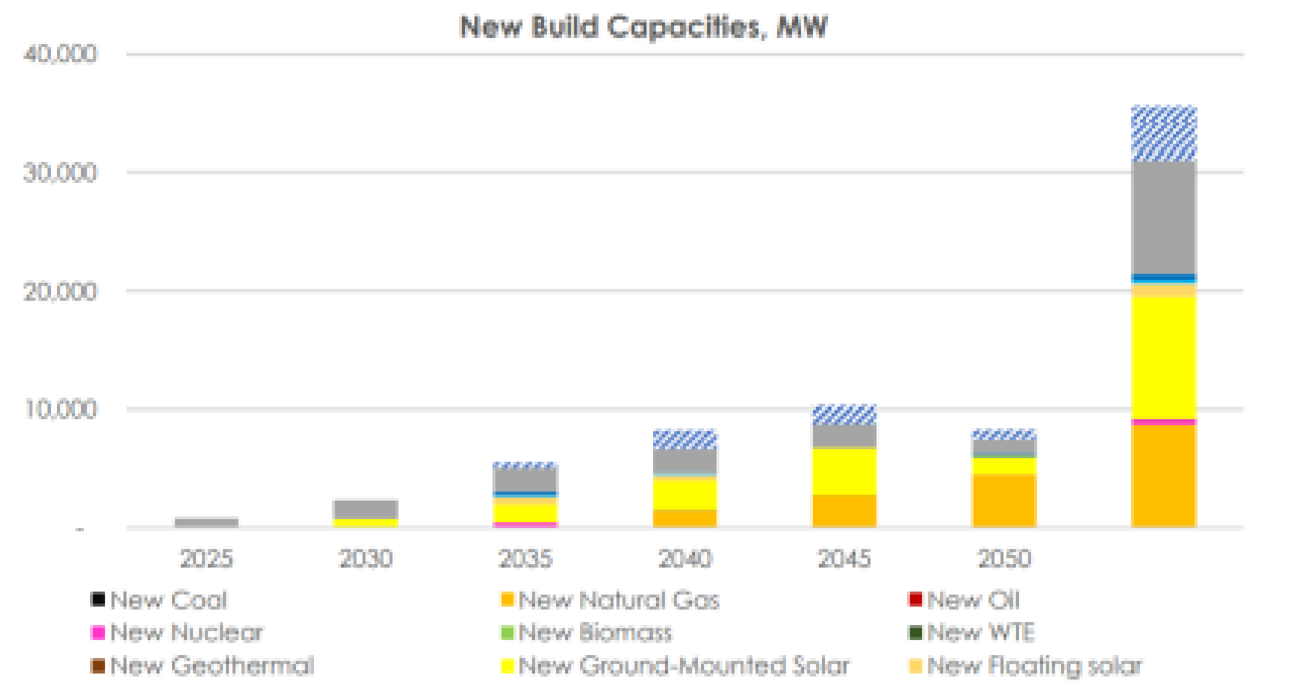

Figure 6.2: CES 1 - Visayas

Data Collection: Mindanao PDP Scenarios

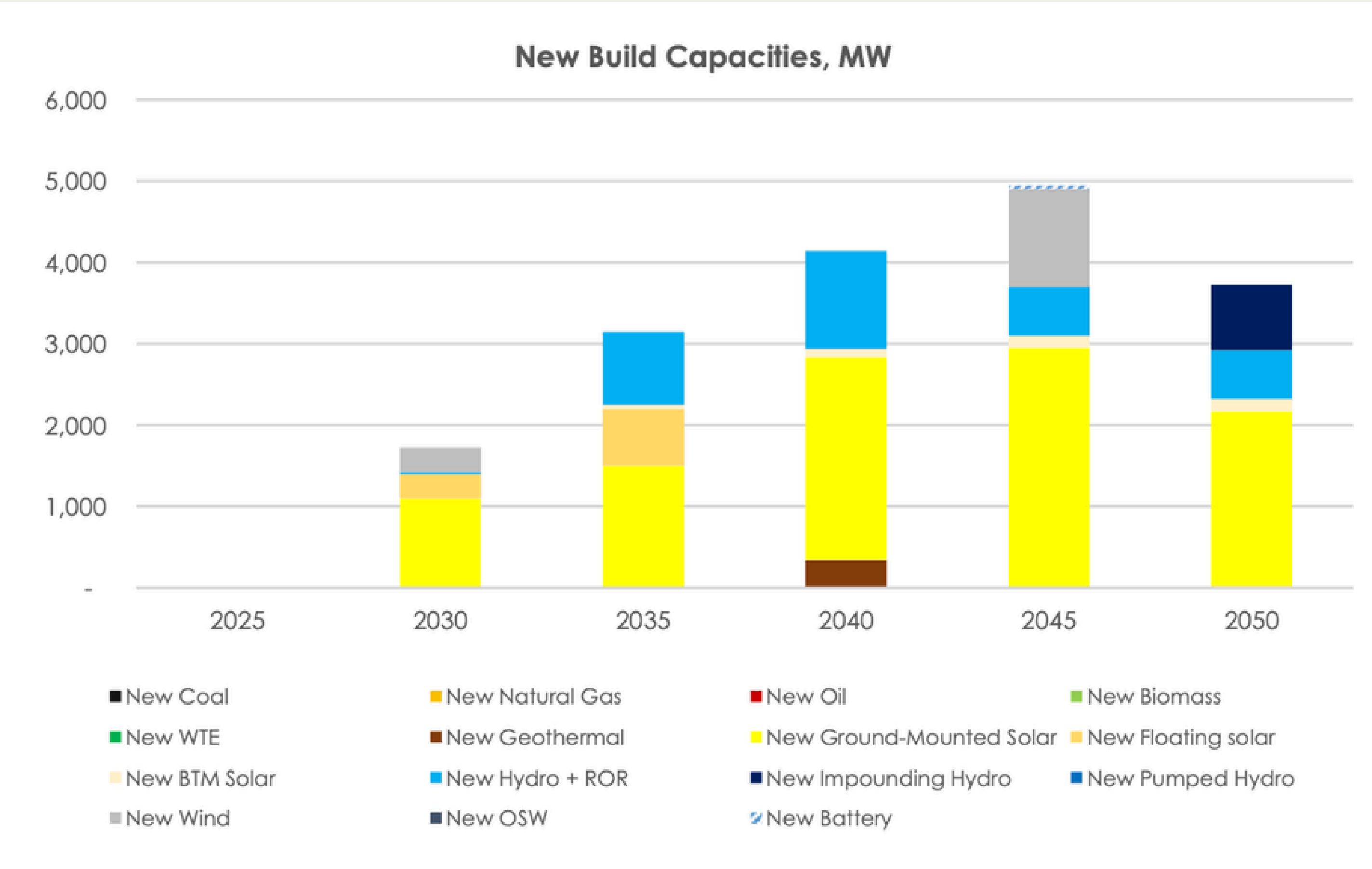

Figure 6.1: Reference - Mindanao

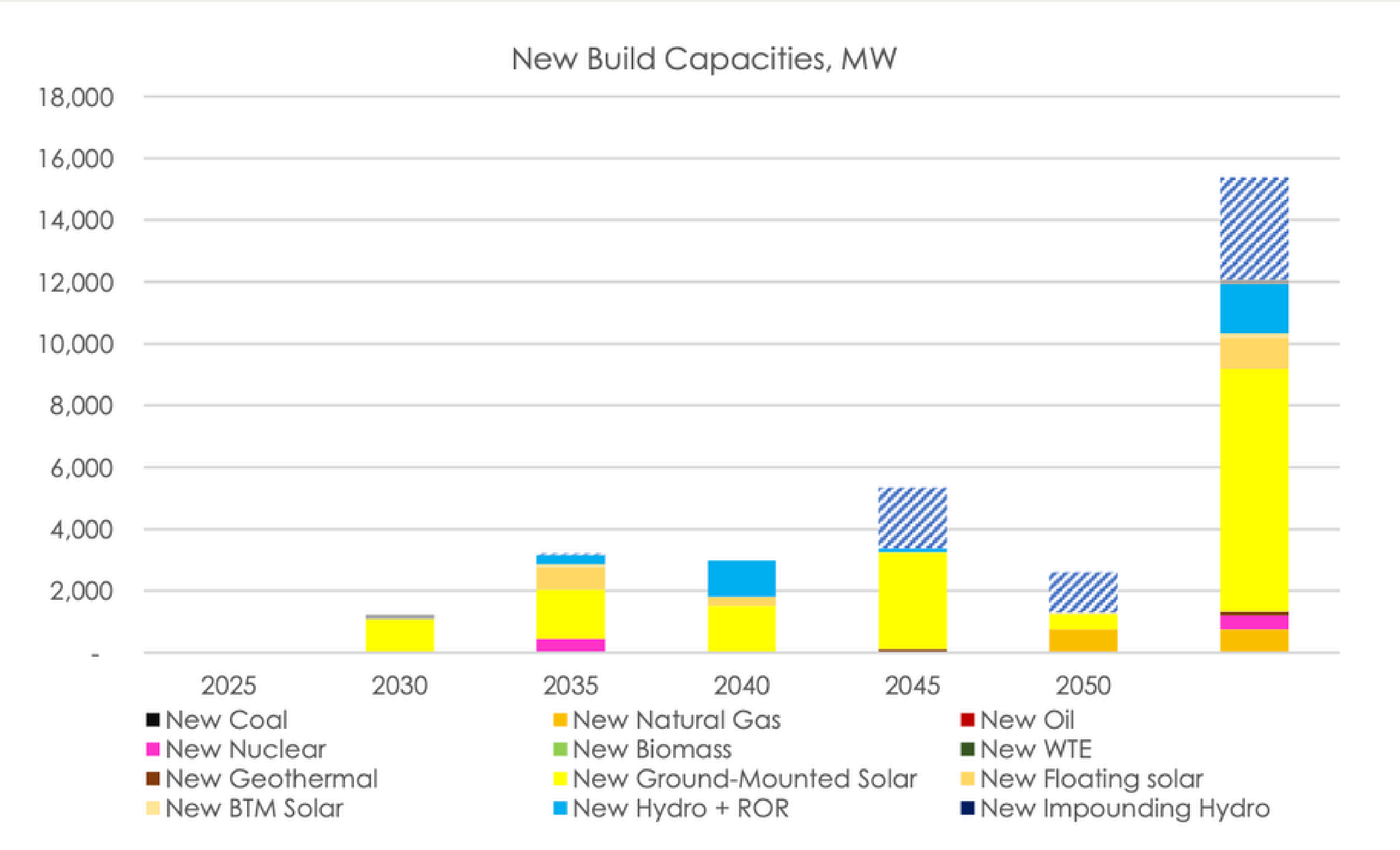

Figure 6.2: CES 1 - Mindanao

Data Collection:
Competitive Renewable Energy Zones

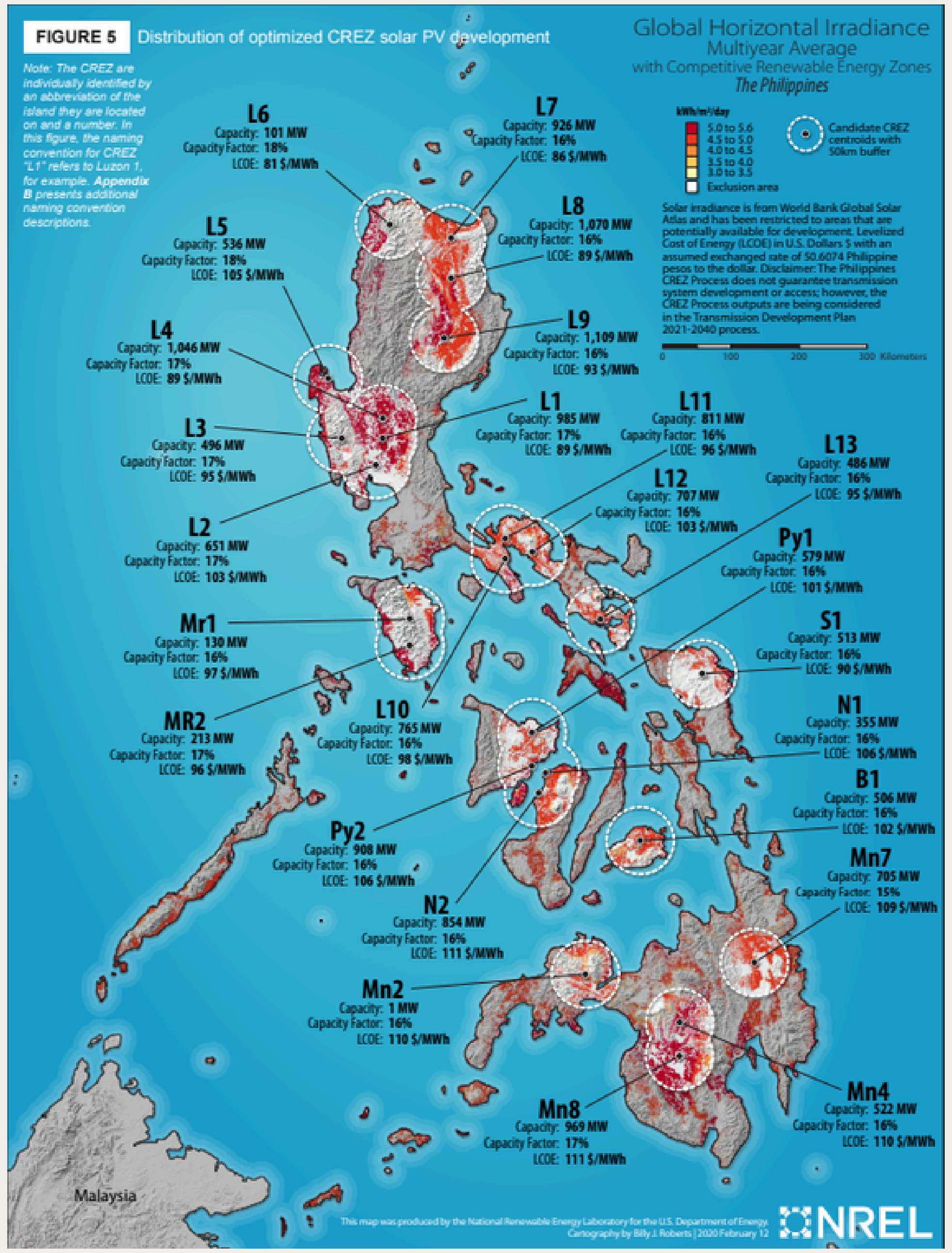

Figure 7.1: Solar CREZ [26]

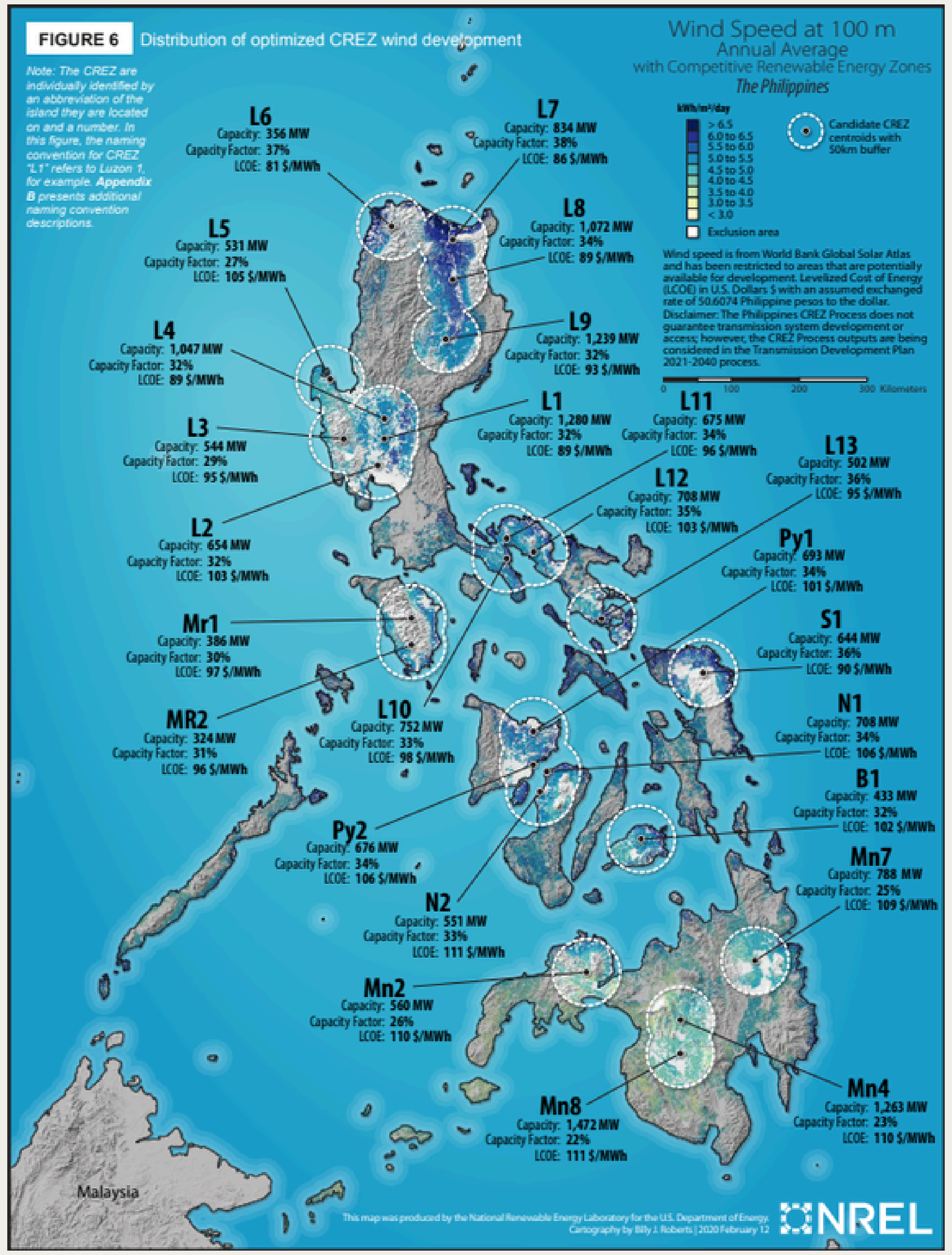

Figure 7.2: Wind CREZ [26]

Data Sources and Parameters

Table 2: Data Parameters and Sources

Data Parameter	Source
Forecasted Demand	Power Development Plan 2023–2050
Build Capacities Per Technology	
Weather Data	Renewables.ninja
Capacity Factor	
CREZ Capacities	Grid Planning and Competitive Renewable Energy Zones (CREZ) in the Philippines
CREZ Coordinates	Renewable Energy Data Explorer

Methodology: Data Processing

Two factors will be considered in developing the composite profile:

Technical Potential (Capacity): Focuses on the weights of each CREZ

$$w_{r,i} = \frac{z_{r,i}}{\sum_{r \in R} z_{r,i}}$$

Where:

$w_{r,i}$	weight of an individual Competitive Renewable Energy Zone of one source (i.e., solar or wind) in a single grid (i.e., Luzon, Visayas, or Mindanao)
$z_{r,i}$	capacity of a single zone (e.g., L1, L2)

Hourly Normalization: Based on the variability of the generation of each source (solar or wind) per hour.

- Solar PV: System Loss = 10%, Tilt = 10 degrees, Azimuth = 180 degrees(South), Capacity = 1 MW [20]
- Wind: Hub Height = 100 m, Turbine Model = Vestas V90 2000, Capacity = 1 MW [20]

Methodology: Data Processing

Finally, using the weights of each CREZ and the hourly data from Renewables.ninja, the hourly composite of each RE will be computed.

$$g_{\text{composite},p,h} = \sum_{r \in R} (g_{r,h,i} \times w_{r,i})$$

Where:

$g_{\text{composite},p,h}$	hourly composite value for a specific source (solar or wind) in one grid (Luzon, Visayas, or Mindanao)
$g_{r,h,i}$	hourly capacity factor based on weather conditions for a specific source (solar or wind) in one grid (Luzon, Visayas, or Mindanao)

Methodology: Model Building

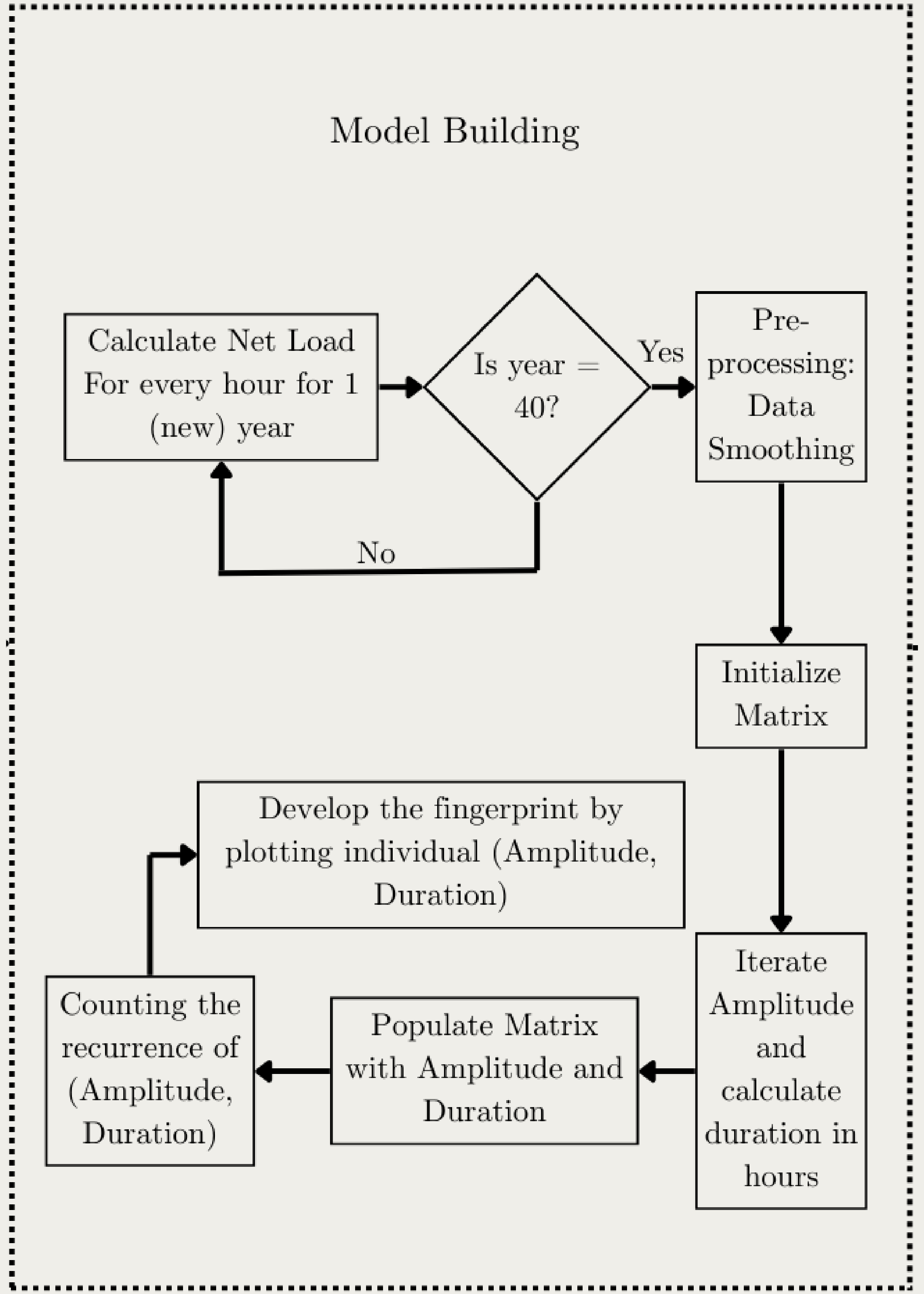

Figure 3: Model Building

Net Load Calculation

$$n_{y,h} = d_{\text{pdp}} - [(g_{\text{composite},w,t} \times c_{\text{pdp},w}) + (g_{\text{composite},s,t} \times c_{\text{pdp},s})]$$

Where

$n_{y,h}$	net load for the specific weather-year for 1 hour
d_{pdp}	projected demand for one year (i.e. 2030, 2035, . . . , 2050) from the PDP
$g_{\text{composite},w,h}$	hourly composite value for for wind
$c_{\text{pdp},w,h}$	the installed capacity target for solar set in PDP
$g_{\text{composite},s,h}$	hourly composite value for for solar
$c_{\text{pdp},s,h}$	the installed capacity target for wind set in PDP

Methodology: Model Building

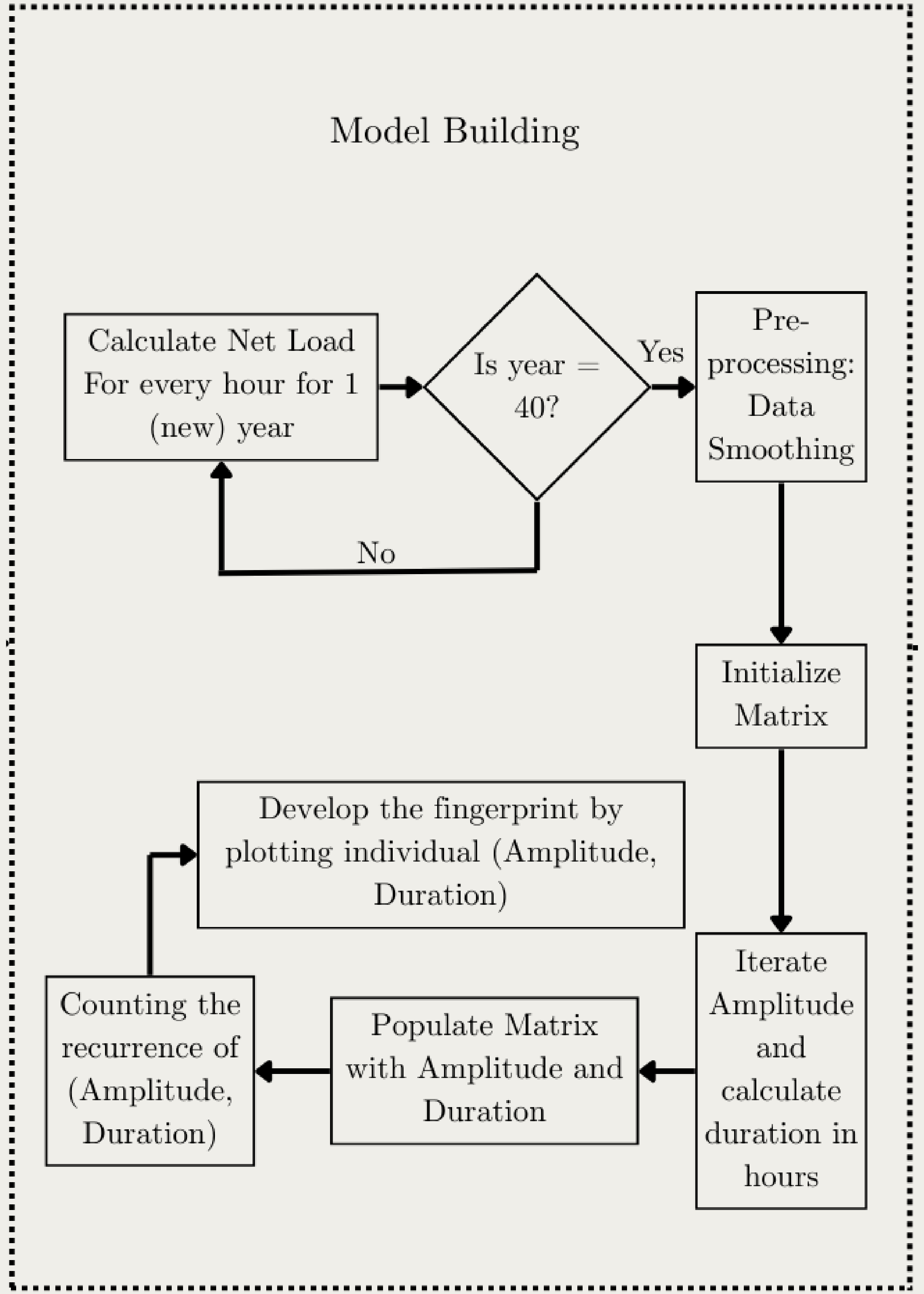

Figure 4: Model Building

Recurrence Matrix and Heatmap Development

Data Smoothing

- 12-hour rolling average will be applied to the hourly load profile

Event Counting

- Matrix will be populated by identifying continuous net-load periods
- adapts the modified event counting by Gorranson [26] where the algorithm will count each net-load as both its maximum duration and as each discrete duration between zero and the maximum

Methodology: Model Building

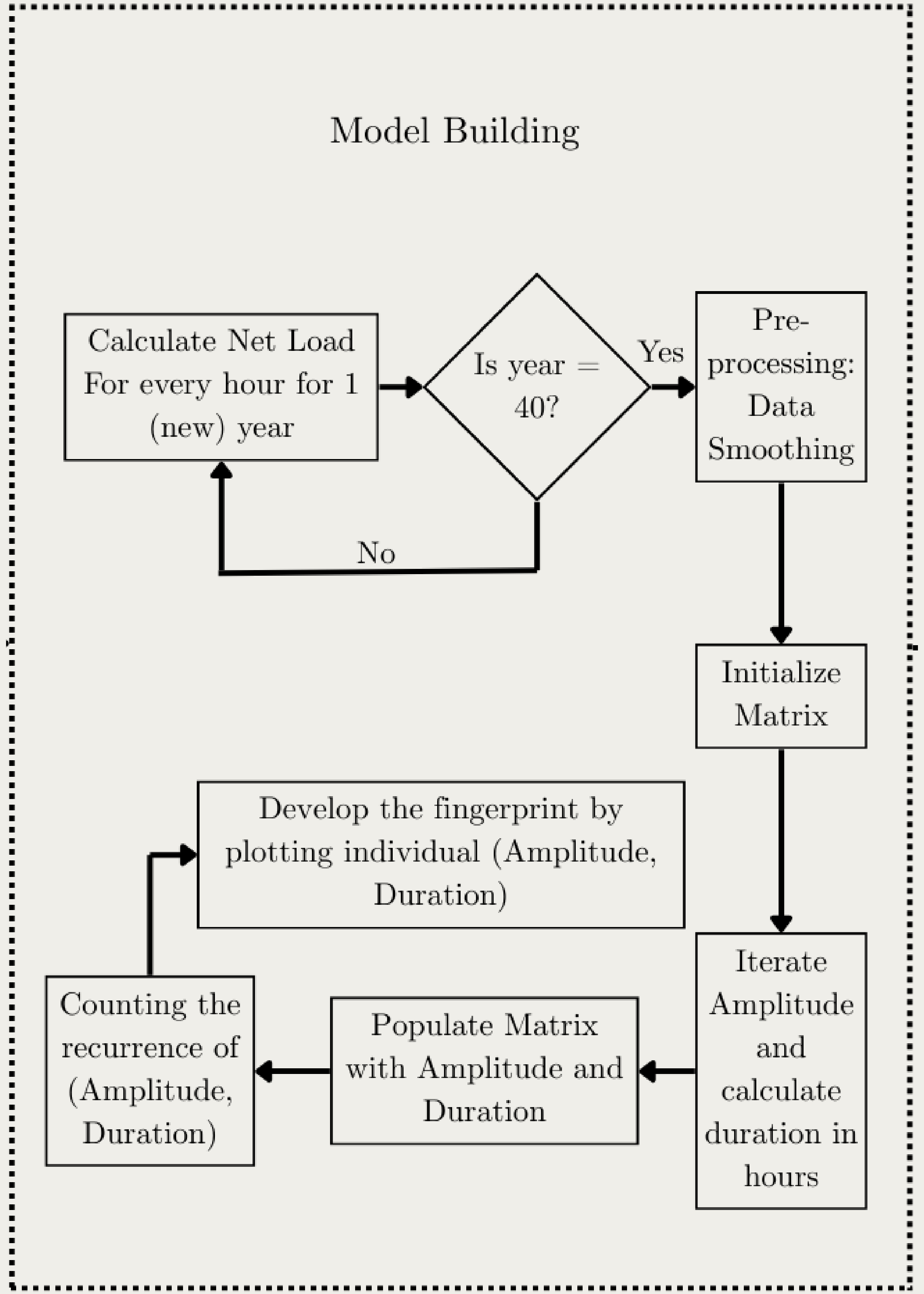

Figure 4: Model Building

Sample Heatmap Recurrence

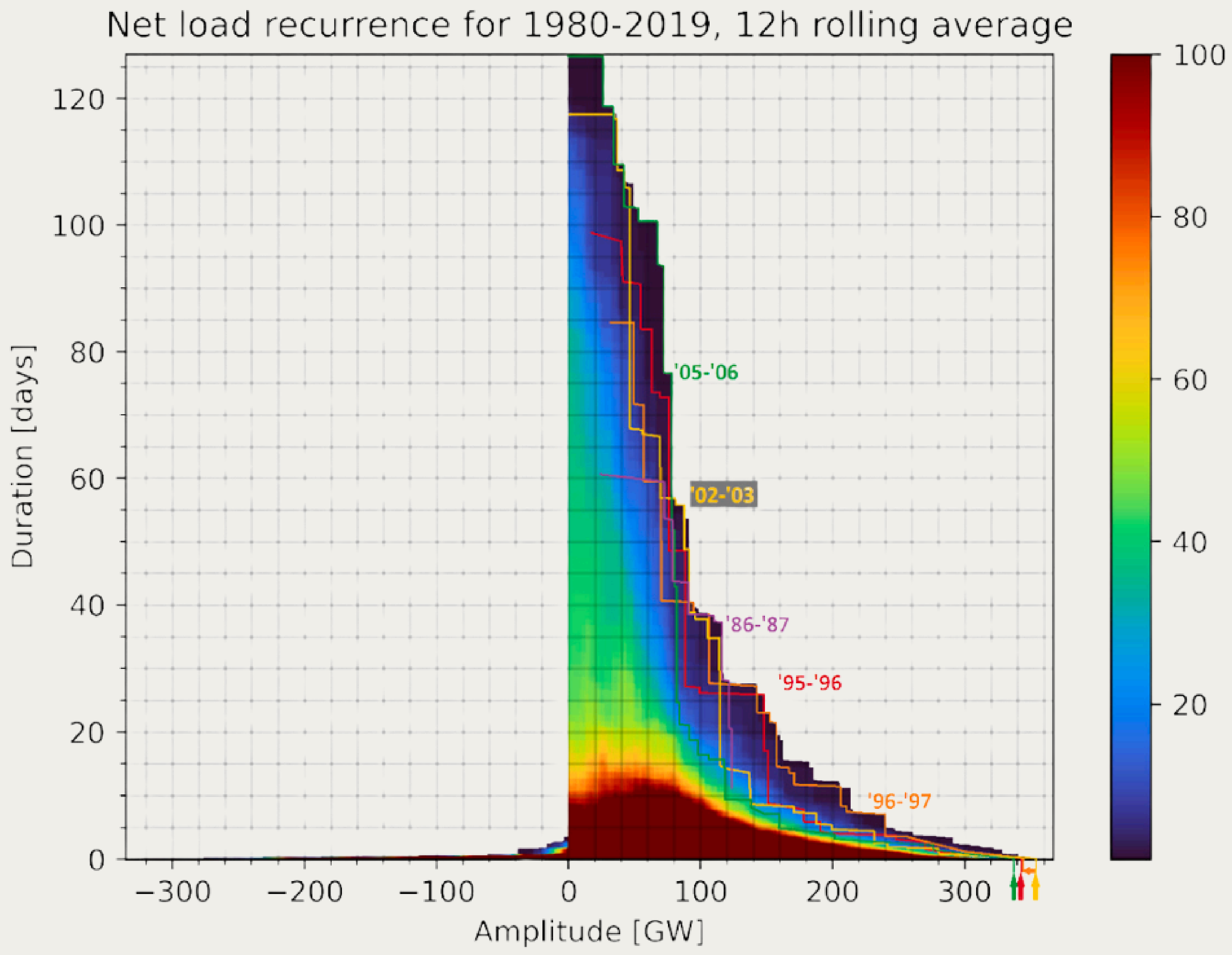

Figure 5: Heatmap of the Recurrence [7]

Methodology: Simulation

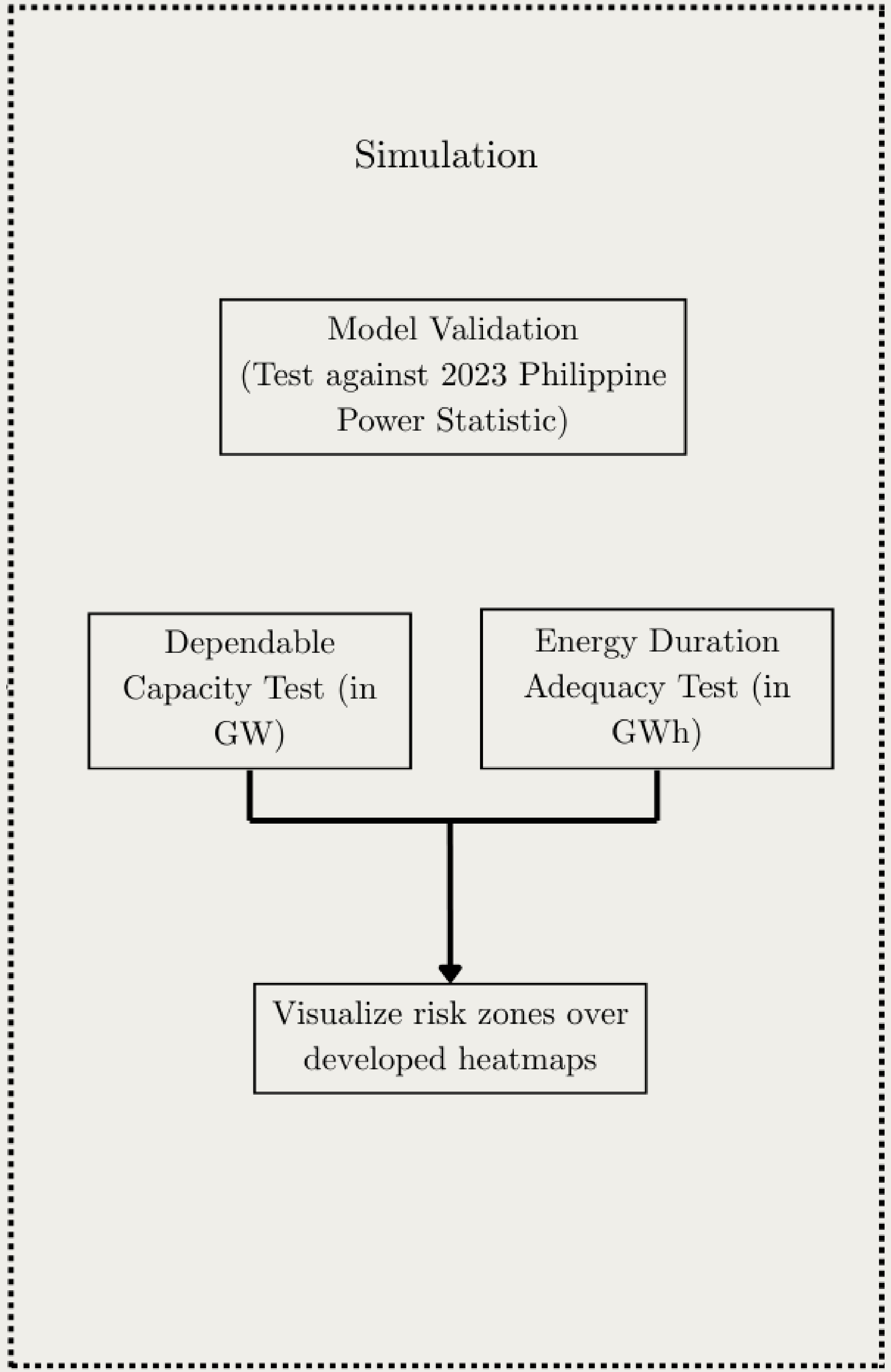

Figure 6: Simulation

Dependable Capacity Test

- Determines the sufficiency of capacity targets to cover the drops in renewable generation

Energy Duration Adequacy Test

- Determines the sufficiency of batteries (in GWh) to outlast a continuous renewable deficiency
- Curve will be developed

$$\tau_{\text{limit}}(x) = \frac{c_{\text{pdp,batt}} \times t_{\text{rated}}}{x}$$

τ_{limit}	maximum sustainable duration (y - axis value); also defines the duration curve
x	net load amplitude (x - axis value)
$c_{\text{pdp,batt}}$	Total Installed Battery Capacity in GW
t_{rated}	Rated Duration Assumption, in this case – 4 hours

Visualization

Capacity Risk Zone

- The right of the vertical axis (red) will be called Capacity Risk Zone where the PDP lacks in dependable capacities.

Energy Risk Zone

- The values above the curve (orange) is the Energy Risk Zone where the batteries cannot handle such long-term events.

Safe Zone

- The area between the curve and the vertical axis (green) are the events that the dependable capacity targets can handle easily.

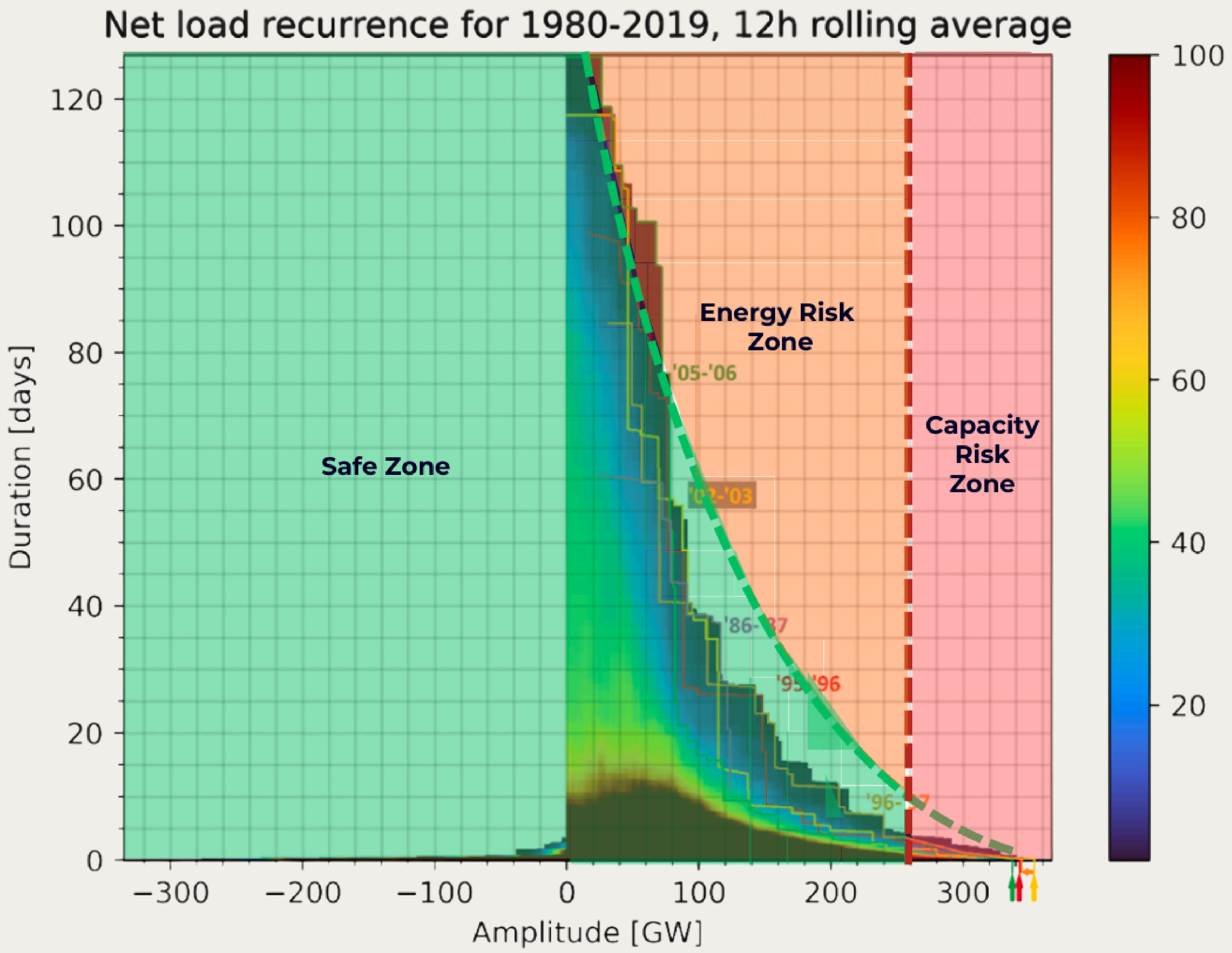

Figure 7: Figure 3 from Ullmark et al [7] with "Capacity Risk Zone", "Energy Risk Zone", and "Safe Zone" Determined

Preliminary Work

Data Gathering

The researchers have completed substantial preliminary work including weather data acquisition, and PDP scenario compilation.

Weather Data

- The weather data for this project has been obtained from Renewables.ninja. Specifically, 16 years (1985-2000) of weather data have been extracted for 10 of the 25 CREZ in Luzon, Visayas, and Mindanao grids designated by DOE.

PDP Reference Scenario, Clean Energy Scenario 1, Clean Energy Scenario 2

- Capacity expansion schedules for Luzon, Visayas, and Mindanao grids have been compiled from the PDP for six planning horizons: 2025, 2030, 2035, 2040, 2045, and 2050

Project Schedule and Deliverables

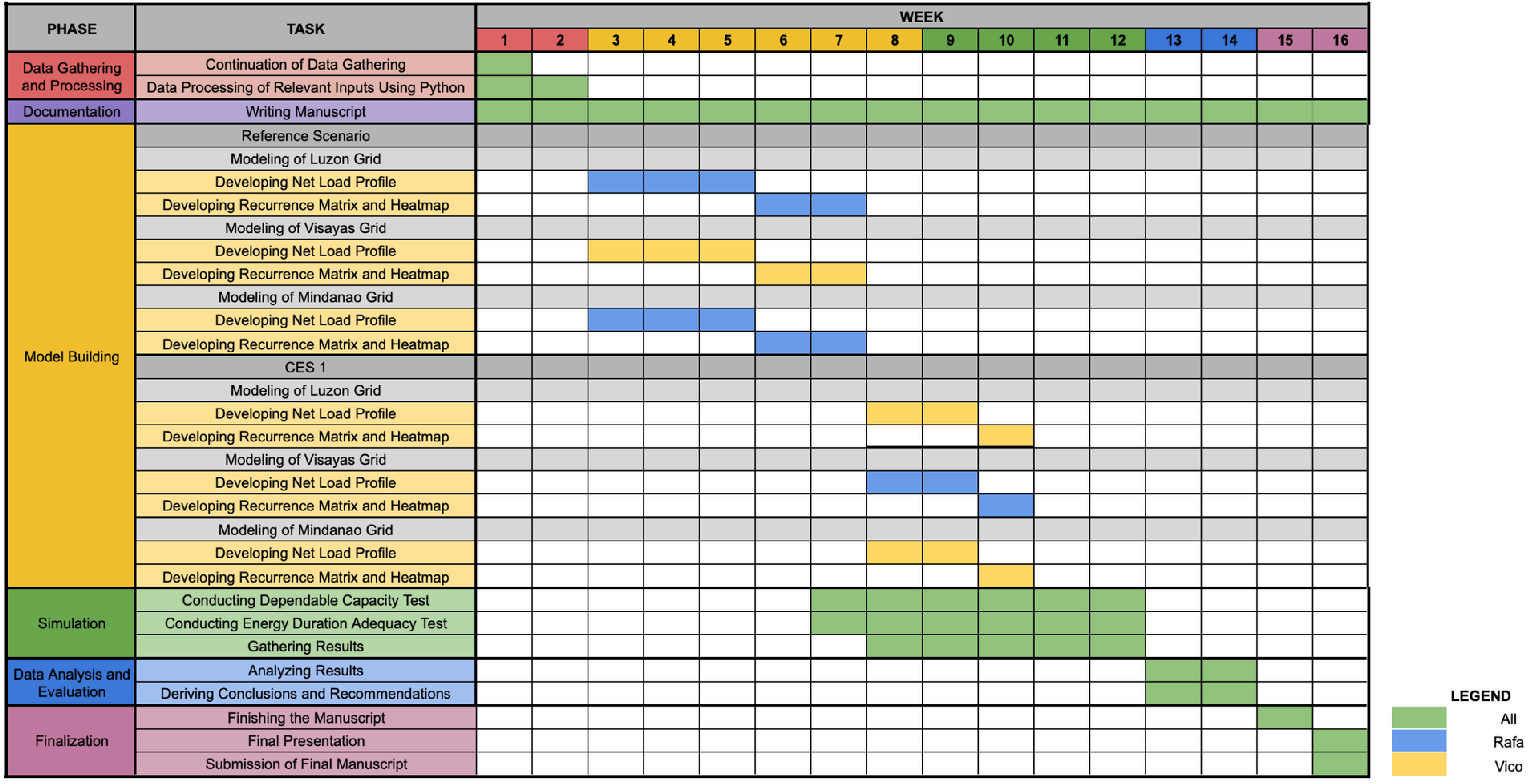

Figure 8: Gantt Chart

Project Schedule and Deliverables

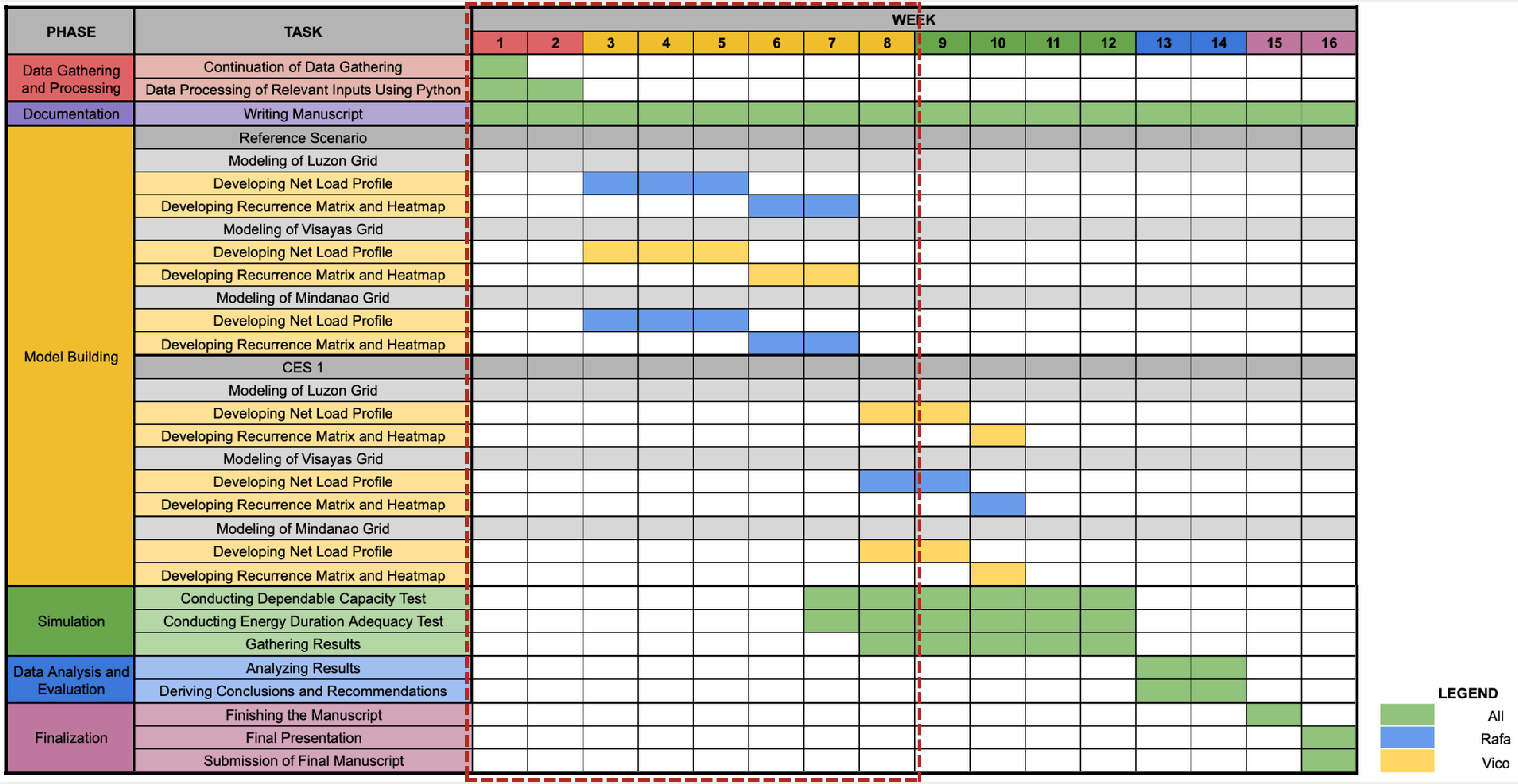

Figure 8: Gantt Chart

Project Schedule and Deliverables

Halfway-point Deliverables

This project aims to accomplish the following mid-point deliverables by Week 8:

1. Gather the data needed for Luzon, Visayas, and Mindanao grids.
2. Process the relevant inputs using Python.
3. Develop net load profile of the grids individually for Reference Scenario.
4. Develop recurrence matrix and heatmap of the grids individually for Reference Scenario.
5. Start developing the net load profile of the grids individually for CES 1.

Project Schedule and Deliverables

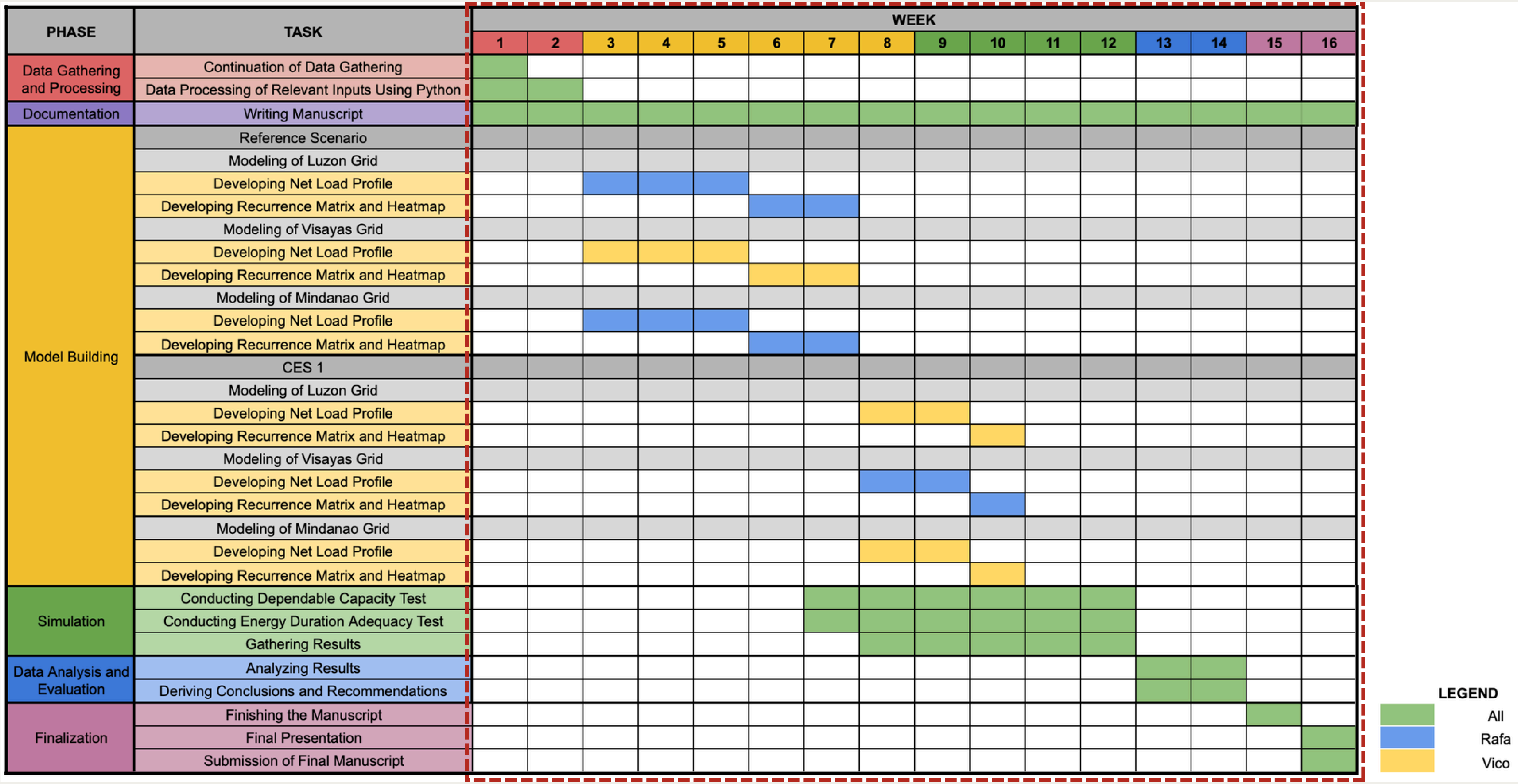

Figure 8: Gantt Chart

Project Schedule and Deliverables

Final Deliverables

By the end of the 16th week, the project is aiming to complete the following deliverables:

1. Repository containing all data inputs and output files generated for every established model.
2. Simulation results to analyze, identify, and quantify any adequacy gap between the PDP's planned capacity and the system's reliability requirements.
3. Slide deck and poster for final presentation.
4. Short paper and appendices that document the project specifics.

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Thank You!